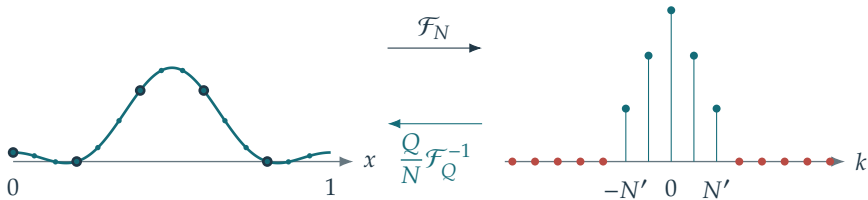


samples on a finer grid

$$|\widehat{f}^N| \longrightarrow |\widetilde{f}^Q|$$



zero padding to length Q

$$N = 2N' + 1, \quad \widetilde{f}_k^Q = \widehat{f}_k^N \quad (|k| \leq N'), \quad 0 \text{ otherwise}$$